AI ACADEMY · LEVEL 4 · GRADE 9 · WEEK 14 · STUDENT LESSON

Scores and Calibration Intuition

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Responsible Plant Classifier

Before you start

AI-3 Week 2: probability is not a guarantee for one outcome.

Plan time to read, investigate and explain. Keep predictions separate from observations.

STUDENT LESSON · 1 OF 18

Your mission

> Start with the target: Interpret calibration using groups of forecasts. Try the recall before opening notes.

Your learning target

I can interpret **calibration** using groups of forecasts.

I can compare average predicted probability with **observed frequency**.

I can distinguish **calibration** from individual correctness.

Remember before reading

Close your notes first. A rough first answer helps us choose the right starting point; it is not a **score** for your ability.

Without opening the old lesson, explain one key idea from Train a Classifier. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Decision Trees. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Leakage. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the **evidence** shows was wrong.

A model **score** can rank **prediction**s without matching the real frequency of correct outcomes.



Your target

LEARNING TARGET Twenty cases receive **score** .7 and fourteen are correct. Explain what the observed rate suggests and what it does not establish about another **score** group.

Builder name and experiment date:

Readiness for the connected explanation

Does a **score** necessarily represent a probability?

This week's end goal

By the end, I can explain **score**s and **calibration** intuition using a traced example and a stated limitation.

STUDENT LESSON · 2 OF 18

Notice the evidence

> Investigate the supplied Scores and Calibration Intuition case. Separate observations, given facts and assumptions before choosing an explanation.

PICTURE ENTRY

Notice → Predict → Explain

Begin with visible **evidence**; do not explain more than the picture and information support.



Start with this puzzle

MYSTERY The team must use **score**s and **calibration** intuition to solve a model-building problem. What information do they have? What important **evidence** is still missing?

Name three visible details without interpreting them.

Predict the data, model, or evaluation decision being made.

Identify one hidden assumption that could change the result.

State what **evidence** would confirm or reject your **prediction**.

My **prediction** and supporting **evidence**:

A closer explanation

If many comparable cases receive probability .8, a calibrated model should produce positives in roughly 80 percent of those cases over adequate data.

It does not mean every case is 80 percent correct or that exactly eight of each next ten must be positive. **Calibration** is assessed across groups.

Use fictional classroom data throughout this investigation. Keep the task, units, reference labels and data split visible as you reason.

Earlier weeks supplied the model-building **process**; this week examines **calibration** as a group property.

STUDENT LESSON · 3 OF 18

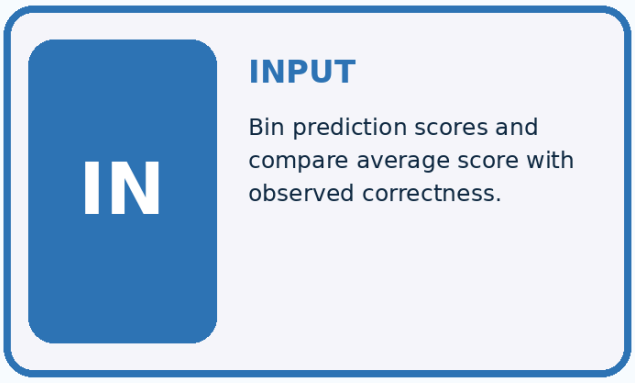
See how it works

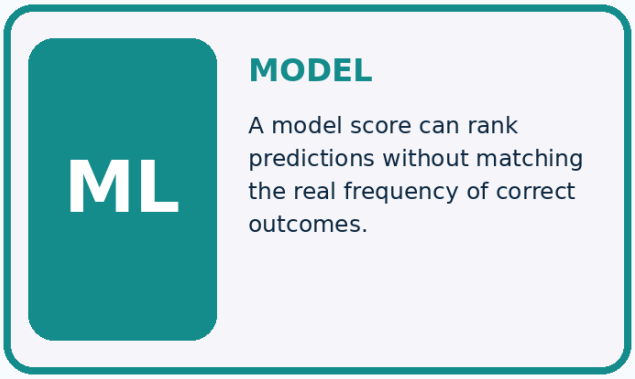
> Follow the mechanism for Scores and Calibration Intuition. Connect each labelled input, deciding step and result.

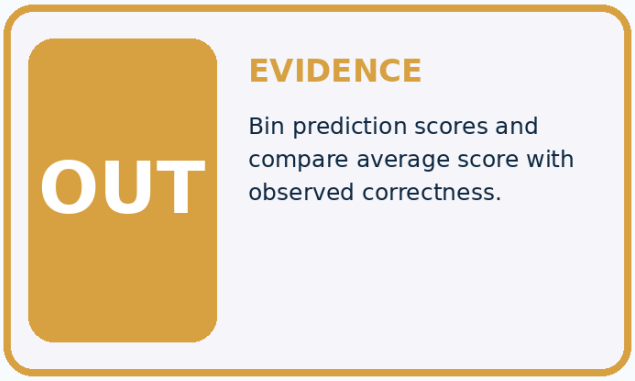
ML MECHANISM

Input → Model → Evidence

Trace the information, fitted behavior, and evaluation proof separately.







Question to explain

QUESTION Bin **prediction score**s and compare average **score** with observed correctness.

MODEL IDEA A model **score** can rank **prediction**s without matching the real frequency of correct outcomes.

PROOF NEEDED Use an unseen case, visible calculation, error analysis, and a stated limitation.

How the mechanism works

UNDER THE HOOD For this example, each **score** estimates the probability that the selected label is correct. Ten **prediction**s have **score**s near 0.8; six are correct, so observed correctness is 6/10=**60%**.

If instead **score**s mean probability of the spotted class, compare them with the fraction of reference-spotted leaves.

Draw a fresh data → model → **prediction** → evaluation trace:

A closer explanation

In a fictional held-out check, ten cases all have predicted probability .2 and two are positive. Another ten all have probability .8 and six are positive.

Average **prediction**s are .2 and .8; observed frequencies are 2/10=.2 and 6/10=.6. The second group is overconfident for positives in this sample.

```
groups = [(0.2, 10, 2), (0.8, 10, 6)]
for score, count, positives in groups:
    if count == 0:
        print("No evidence in this group")
    else:
        print(score, positives / count)
# Compare each score with its observed fraction.
# A group fraction is not an individual guarantee.
```

Trace the calculation on paper before running this optional Python check. Use the stated fictional **input**s and compare intermediate values. The code checks this example; your independent workbook case uses different **evidence**.

Connect the picture and the explanation

Point to the **input**, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

STUDENT LESSON · 4 OF 18

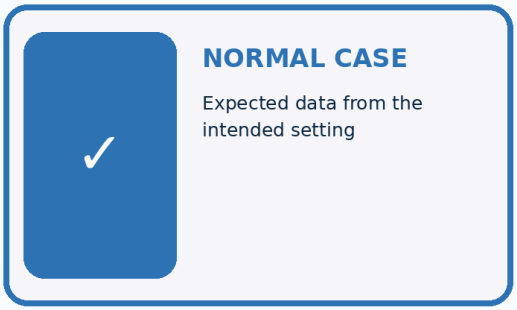
Compare the cases

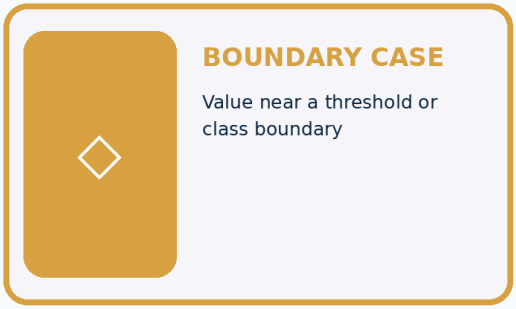
> Compare the supplied cases for Scores and Calibration Intuition. Identify the detail that changes the conclusion.

EVALUATION DESIGN

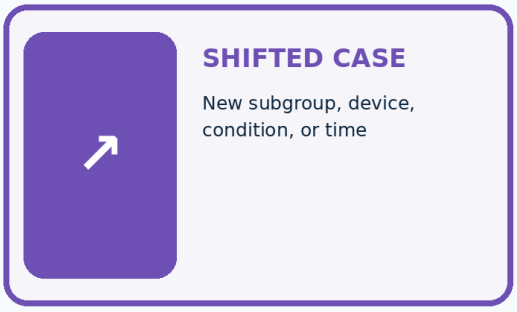
Four Tests, Four Kinds of Evidence

A system is not understood until it survives more than one comfortable example.









NORMAL Expected data and expected behavior.

Keep this boundary clear

BOUNDARY Value close to a threshold, ambiguous label, or uncertain **prediction**.

MISSING Necessary field, label, source, or measurement is unavailable.

SHIFTED New lighting, species, device, subgroup, or time condition.

RELEASE RULE Passing one case never proves generalization. Predefine expected behavior for all four categories.

Which case most threatens today's claim, and why?

Change one thing in the pictured case

Change one thing

Change: The 0.2 group has two spotted cases instead of four.

Predicted result

Prediction: Its observed fraction becomes 0.2 and matches the supplied **score** in this group.

Why the result follows

Reason: **Calibration** compares stated probabilities with group outcome frequencies.

STUDENT LESSON · 5 OF 18

Words that unlock the idea

> Use the vocabulary of Scores and Calibration Intuition to explain a concrete decision, rather than only memorising definitions.

VOCABULARY

Words You Can Calculate and Defend

A definition matters when it lets you interpret a new dataset, model, or result.


1
SCORE
model value used for ranking or confidence
2
CALIBRATION
scores align with observed frequencies
3
BIN
group of predictions with similar scores
4
RELIABILITY
agreement between score and outcome rate

Term	Precise meaning	My ML example
score	model value used for ranking or confidence	
calibration	scores align with observed frequencies	
bin	group of predictions with similar scores	
reliability	agreement between score and outcome rate	

MISCONCEPTION TO REPAIR A **score** of 0.9 means the **prediction** has exactly a 90% chance of being correct.

Repair the claim using a definition, calculation, or test record:

Words used in the closer explanation

Word	Meaning
Calibration	Agreement between predicted probabilities and outcome frequencies across comparable groups
Bin	A declared interval grouping predictions
Observed frequency	Positive outcomes divided by cases in a group
Reliability diagram	A plot comparing group average prediction with observed frequency

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STUDENT LESSON · 6 OF 18

Follow a worked example

> Follow the worked trace for Scores and Calibration Intuition, predict a step, then compare with its explanation.

WORKED EVIDENCE

Reason Step by Step

Predict first. Then check each step. A correct answer without visible **evidence** cannot be audited.

Four-step worked evidence for Week 14: Scores and Calibration Intuition

STEP	EVIDENCE AND REASONING
1	Each score estimates the probability that the selected label is correct. Ten predictions have such scores near 0.8.
2	Only six are correct, so observed correctness is 6 ÷ 10 = 60%.
3	The score ranking may still help, but 0.8 should not be stated as an 80% guarantee.
4	Use bins across many cases and design an uncertain zone requiring human review.

Trace these corrected steps. Keep the stated metric, assumptions and interpretation when you change the case.

Step 1

STEP 1 Each **score** estimates the probability that the selected label is correct. Ten **prediction**s have such **score**s near 0.8.

Step 2

STEP 2 Only six are correct, so observed correctness is 6 ÷ 10 = **60%**.

Step 3

STEP 3 The **score** ranking may still help, but 0.8 should not be stated as an **80%** guarantee.

Step 4

STEP 4 Use bins across many cases and design an uncertain zone requiring human review.

CHECK Verify the denominator, unit, split, class definition, and data version before interpreting a **score**.

Recalculate, retrace, or restate the **evidence** independently:

A closer explanation

Calibration as a group property at a glance

PART	EVIDENCE OR ACTION	INTERPRETATION
Low probability group	Average .2	Observed 2 of 10
High probability group	Average .8	Observed 6 of 10
Compare groups	Prediction versus frequency	Keep group sizes visible

Plot average prediction on the horizontal axis and observed frequency on the vertical axis. The two points are (.2,.2) and (.8,.6). Agreement lies along the diagonal. Use the average score within a bin, not automatically the bin midpoint. Small groups make frequencies uncertain.

Which worked group overstates positive frequency and by how much? Explain the steps or evidence behind your response.

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Scores and Calibration Intuition

Treat **calibration** as a comparison across groups of probability **score**s.

The facts we will use

Supplied facts

Toy classifier **output**s p(spotted)=0.8 on ten cases; eight are actually spotted. Another ten have p=0.2; four are spotted.

Supplied facts

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

AI ACADEMY / AI-4/14

Scores and Calibration Intuition

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Group scores

Keep the ten 0.8 cases and ten 0.2 cases separate.



2. Compare frequencies

Observed spotted fractions are 8/10=0.8 and 4/10=0.4.



3. Interpret gap

The 0.8 group matches here; the 0.2 group understates the observed fraction by 0.2.

CHANGE ONE THING

The 0.2 group has two spotted cases instead of four.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Group scores	Keep the ten 0.8 cases and ten 0.2 cases separate.
2. Compare frequencies	Observed spotted fractions are 8/10=0.8 and 4/10=0.4.
3. Interpret gap	The 0.8 group matches here; the 0.2 group understates the observed fraction by 0.2.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

Why the result follows

Read the picture in order. The 0.8 group matches here; the 0.2 group understates the observed fraction by 0.2. The arrows show which recorded **input** or intermediate result each decision uses.

Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

What this example does not establish

Keep the stated limit

Ten cases per group provide noisy **evidence**; the exercise does not establish **calibration** on every **score** or setting.

Keep the stated limit

Read the labelled picture: Does a calibrated 0.8 **score** mean every such leaf is spotted? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Keep the stated limit

Change one thing: The 0.2 group has two spotted cases instead of four. Predict the result before checking the explanation. What remains the same?

STUDENT LESSON · 7 OF 18

Find and repair a mistake

> Find the unsupported step in the Scores and Calibration Intuition claim and explain a repair using the supplied evidence.

Find the first error

ERROR DETECTIVE

Compare → Diagnose → Repair

Do not patch the answer. Find the earliest unsupported assumption, data problem, or experiment error.

!

COMPARE • DIAGNOSE • REPAIR

Tempting claim: A score of 0.9 means the prediction has exactly a 90% chance of being correct. Inspect the dataset, split, calculation, error type, subgroup, or experiment record; identify one testable cause; change one factor; rerun the same evidence.

Evidence record	Expected	Actual	Likely cause / next test
normal case	passes		
boundary case	defined behavior		
missing case	safe fallback		
shifted case	measured limit		

CLAIM TO CHALLENGE A **score** of 0.9 means the **prediction** has exactly a 90% chance of being correct.

Diagnose one cause and name the cleanest next experiment:

A closer explanation

Bin boundaries, group sizes and dataset coverage affect the picture. Do not fit adjustments on the final test or claim certainty from a small bin.

Real distributions can change, and a calibrated model on one population may not remain so elsewhere.

> **What is wrong with calibrating and reporting success on the same final-test records?**

STUDENT LESSON · 8 OF 18

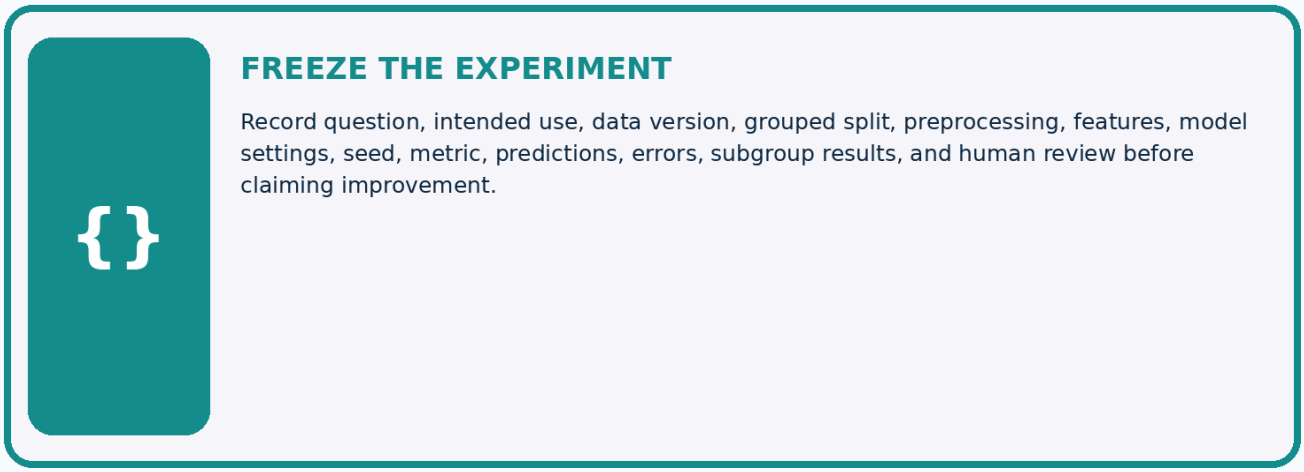
Practise together

> Practise Scores and Calibration Intuition with support, then explain the deciding step in your own words.

ML STUDIO

Specify Before You Run

A trustworthy result can be reproduced from the recorded data, code, settings, and evaluation rule.



Experiment record

> **Question to explain**
>
> QUESTION + intended use + success threshold

DATA version + grouped split + leakage check

CODE/pre**process**ing + features + model + settings + seed

METRICS + positive class + units + denominators

PREDICTIONS + errors + subgroup slices + limitations

HUMAN reviewer + release, revise, limit, or stop decision

Write the minimum record needed to reproduce today's result:

A closer explanation

If every case gets .5 and exactly half are positive, overall **calibration** can look good while the model cannot rank easy and difficult cases. Conversely, useful ranking can coexist with poor probabilities.

Report discrimination and **calibration** separately. Fit any **calibration** adjustment on development data reserved for that purpose, then evaluate on separate data.

Practise with less help: Locate the discrepancy

Use workbook W2 for the connected example "**Calibration** as a group property". First fill the **input** and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan	Do	Check
What is given? Which rule or procedure applies?	Record each intermediate step.	Does the result follow? What remains unknown?

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Check your reasoning

Use the worked case for **Calibration** as a group property. Proposed explanation: “A final answer is enough”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Answer on your own before discussion. If you are unsure, say which step you need help with.

Explain the picture

Does a calibrated 0.8 **score** mean every such leaf is spotted? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

> **My explanation using a pictured step and the given facts**

This is supported practice. Keep the separately named independent case for an attempt before feedback.

STUDENT LESSON · 9 OF 18

Run the investigation

> Investigate Scores and Calibration Intuition. Record the prediction, actual observation or labelled paper trace, explanation and next test separately.

EVIDENCE LAB

Predict → Run → Calculate → Interpret

Keep the test conditions fixed, record every result, and change only one planned factor.

LAB

PREDICT → RUN → CALCULATE → INTERPRET

Reliability Lab: sort thirty predictions into score bins, calculate observed correctness, and identify overconfident regions.

1. Write the hypothesis and expected **evidence** before running.

2. **Reliability** Lab: sort thirty **prediction**s into **score** bins, calculate observed correctness, and identify overconfident regions.

3. Run one normal, one boundary, one missing, and one shifted case.

4. Calculate the metric from raw counts or row-level errors.

5. Interpret what changed, what did not, and what remains unsafe or unknown.

Case · Prediction · Actual · Calculation / evidence · Meaning / next test

> **Case:** Normal

Prediction: _____

Actual: _____

Calculation / evidence: _____

Meaning / next test: _____

Case: Boundary

Prediction: _____

Actual: _____

Calculation / evidence: _____

Meaning / next test: _____

Case: Missing

Prediction: _____

Actual: _____

Calculation / evidence: _____

Meaning / next test: _____

Case: Shifted

Prediction: _____

Actual: _____

Calculation / evidence: _____

Meaning / next test: _____

Plan → try → check

Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

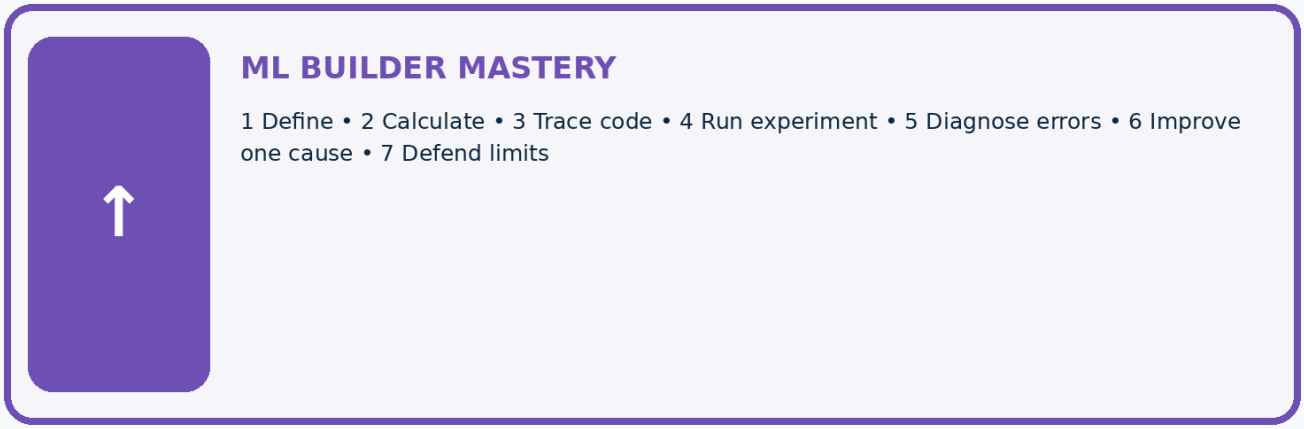
Build your understanding

> Move from recall to tracing, diagnosis and transfer in Scores and Calibration Intuition. Explain why each step is justified.

MASTERY

From Knowing the Word to Owning the Idea

Move upward only when you can produce **evidence** without copying the example.



DEFINE Use all four terms precisely: **score**, **calibration**, bin, **reliability**.

CALCULATE Reproduce the worked **evidence** with units and denominator.

TRACE Explain the data, code or decision path.

DIAGNOSE A **score** of 0.9 means the **prediction** has exactly a 90% chance of being correct.

TRANSFER Solve a fresh dataset or model case.

IMPROVE Change one cause and rerun fixed **evidence**.

DEFEND State intended use, metric, limitation, and human responsibility.

MASTERY RULE A coached calculation becomes mastery only after a different case is completed independently.

STUDENT LESSON · 11 OF 18

Connect your project

> Apply this week's idea to Responsible Plant Classifier. Keep an annotated decision and the evidence supporting it.

CUMULATIVE PROJECT

Responsible Plant Classifier

Every week adds one reviewable artifact to the same **evidence**-based capstone.

RESPONSIBLE PLANT CLASSIFIER

Add an uncertainty band to the Plant Classifier and define what the interface says when scores fall inside it.

THIS WEEK'S CHECKPOINT Add an uncertainty band to the Plant Classifier and define what the interface says when **score**s fall inside it.

Which requirement or risk does this artifact address?

Which dataset, split, code, and metric version does it use?

What normal and shifted cases were tested?

What remains uncertain or weak?

What decision still belongs to a teacher or qualified human?

RESPONSIBLE RELEASE Never use this classroom prototype for medical, pesticide, agricultural, or other consequential decisions.

My **evidence**-linked project decision:

Evidence for your project

For your Responsible Plant Classifier, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

STUDENT LESSON · 12 OF 18

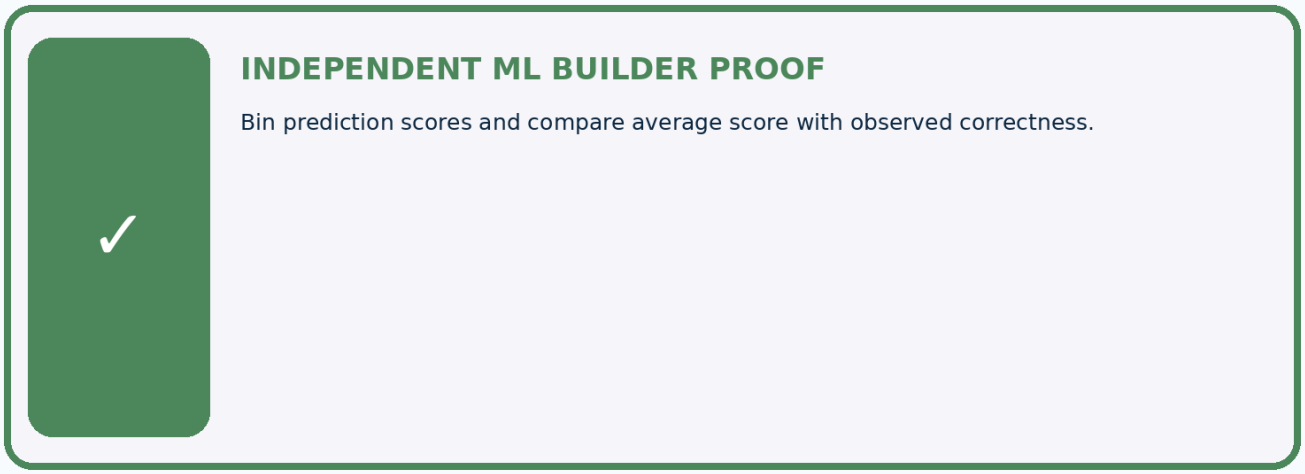
Show what you know

> Use a genuinely new case for independent evidence. Previously discussed examples remain practice; use the separately named independent case before feedback.

INDEPENDENT PROOF

Explain • Calculate • Transfer • Defend

Complete this with a fresh case after guided practice and compare it with the mastery criteria.



What success looks like

REQUIRED PROOF Bin **prediction score**s and compare average **score** with observed correctness.

- ☐ Define the task, feature/target roles, and intended-use boundary.
- ☐ Show the calculation, code trace, or experiment record.
- ☐ Test a normal case and one revealing boundary, missing, or shifted case.
- ☐ Diagnose one failure without changing several factors at once.
- ☐ Name a limitation, fairness concern, and the human-review action.

Independent **evidence** and defense:

My level: ☐ Not yet ☐ With one hint ☐ Independently ☐ I can teach and defend it

Connect to the next lesson

NEXT BRIDGE Carry the experiment **evidence** into: Confusion Matrix.

Study a complete case

> Revisit the complete Scores and Calibration Intuition case for supported practice. Repeated examples do not become fresh assessments.

Week 14 Worked case

Use the supplied details to practise the weekly concept before attempting the independent question. This case extends the illustrated lesson.

The situation

Ten cases receive probability .8 for spotted; eight are spotted.

The question

Interpret **calibration** and its limits.

Worked explanation

8/10 matches .8 in this tiny bin. It illustrates **calibration**, not proof of global **calibration** or individual certainty; test more bins and data.

Explain it in your own words

> Which detail supports the answer, and why does it matter?

Write or draw your reasoning and evidence.

Trace the supplied case visually

Compare a score bin with observed frequency

Evidence or stage	Value or result
Spotted among score-0.8 cases	8/10

Ten cases receive probability .8 for spotted; eight are spotted.

Interpret **calibration** and its limits.

8/10 matches .8 in this tiny bin. It illustrates **calibration**, not proof of global **calibration** or individual certainty; test more bins and data.

STUDENT LESSON · 14 OF 18

Try a new case independently

> Attempt the reserved new case before feedback. Record support; the earlier examples below remain additional practice.

Independent case: Apply independently

Twenty cases all predict .3 with eight positives; ten all predict .7 with seven positives. Give plot points and interpret.

> **My independent explanation, trace and limits**

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Week 14 Independent application

Close the worked explanation. Apply the idea to this different question without copying.

> Among 20 **prediction**s with **score** 0.8, 12 are correct. What is observed accuracy in this group and how does it compare with 0.8?

My answer, with a trace, calculation or supporting **evidence**

> **Write or draw your reasoning and evidence.**

Create and test

Change one relevant detail. Predict the result before testing. Include a boundary or missing-information case when it helps reveal a limit.

My changed **input** and expected result

> **Write or draw your reasoning and evidence.**

Connect to your project

Add an uncertainty band to the Plant Classifier and define what the interface says when **score**s fall inside it.

The **evidence** I will keep

Write or draw your reasoning and evidence.

Your teacher has the independent answer and scoring criteria. Complete the matching workbook week to keep your practice **evidence** together.

Transfer practice from the original programme

Twenty cases receive **score** .7 and fourteen are correct. Explain what the observed rate suggests and what it does not establish about another **score** group.

Explain the mechanism, evidence and one limit.

Use feedback to improve

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed.

A later check uses a changed case, not a copied answer.

STUDENT LESSON · 15 OF 18

Your lesson route

> Use this map to navigate Scores and Calibration Intuition, connect the goals to tasks and keep one evidence record for each attempt.

Start with this question: Treat **calibration** as a comparison across groups of probability **score**s.

By the end, I can	How I will show it
Interpret calibration using groups of forecasts	Explain the deciding step, show a trace or test, and state one limit.
Compare average predicted probability with observed frequency	Explain the deciding step, show a trace or test, and state one limit.
Distinguish calibration from individual correctness	Explain the deciding step, show a trace or test, and state one limit.

Remember → watch and explain → try with help → investigate → try independently → keep project **evidence**. Your teacher may spread this lesson across several classes.

Keep your first **prediction**. Compare it with what the **evidence** shows and explain any change.

STUDENT LESSON · 16 OF 18

Finish and return

> Explain Scores and Calibration Intuition without notes, identify a remaining question and plan a later check with a changed case.

Close your notes. Explain this goal in your own words: Interpret **calibration** using groups of forecasts

My deciding step, evidence and one remaining question

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an **input** or assumption and justify your **prediction**.

Save your workbook **evidence**. At the next review, try a different case before opening your old answer.

STUDENT LESSON · 17 OF 18

Supported practice, repair and portfolio

> Attempt supported questions before using hints or feedback. Review a missing building block, then keep the completed work.

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Treat **calibration** as a comparison across groups of probability **score**s.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? The 0.8 group matches here; the 0.2 group understates the observed fraction by 0.2.

1. Group **score**s

2. Compare frequencies

3. Interpret gap

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Interpret gap The 0.8 group matches here; the 0.2 group understates the observed fraction by 0.2. Read the picture in order.

The 0.8 group matches here; the 0.2 group understates the observed fraction by 0.2. The arrows show which recorded **input** or intermediate result each decision uses.

Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Practice 2: explain the deciding evidence

Changed case · predict first

Changed-case practice: The 0.2 group has two spotted cases instead of four. Which conclusion is supported by the supplied facts?

1. Its observed fraction becomes 0.2 and matches the supplied **score** in this group.

2. This one example guarantees correct results for every future **input**.

3. We can decide the result without examining the changed **input** or the procedure.

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Its observed fraction becomes 0.2 and matches the supplied **score** in this group. **Calibration** compares stated probabilities with group outcome frequencies.

Repair a missing building block

Review Train a Classifier (ai-4/13). Goal: Connect **training**-only scaling and neighbour **prediction** in one pipeline. Short practice: Which pipeline record would help another team reproduce the result?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Decision Trees (ai-4/11). Goal: Fit a root question and leaf **prediction**s from **training** examples. Short practice: Which **input** distinguishes the original and changed tree?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Leakage (ai-4/8). Goal: Choose a split unit that matches the intended **prediction** claim. Short practice: How many independently new plant identities are in the original test?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Predict, run and debug

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual **output** or error, change one **input**, then run again.

Keep code, **prediction**s, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

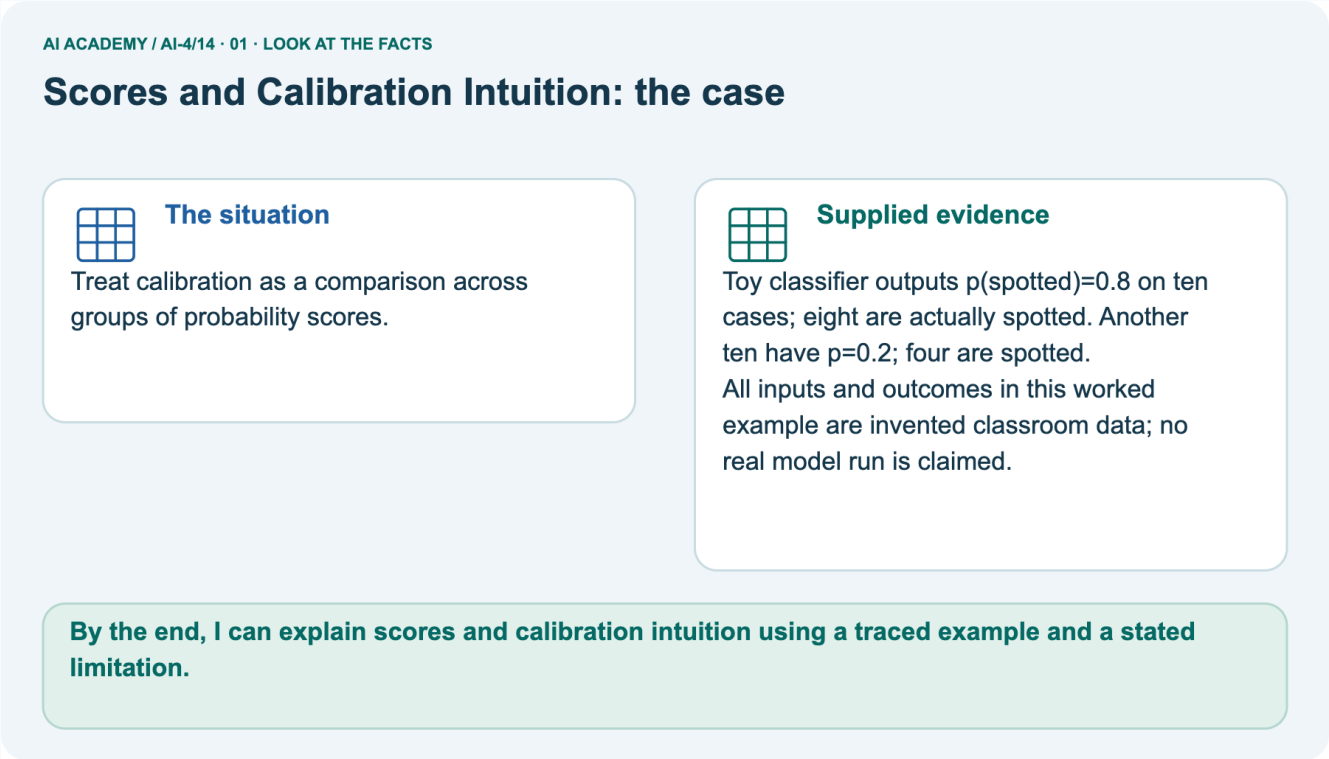
Drafts stay in this browser when storage is available; clear drafts on a shared device.

STUDENT LESSON · 18 OF 18

Visual concept journey

> Explain each Scores and Calibration Intuition graphic using its labels, deciding evidence and relevant limitation.

01. Scores and Calibration Intuition: the case



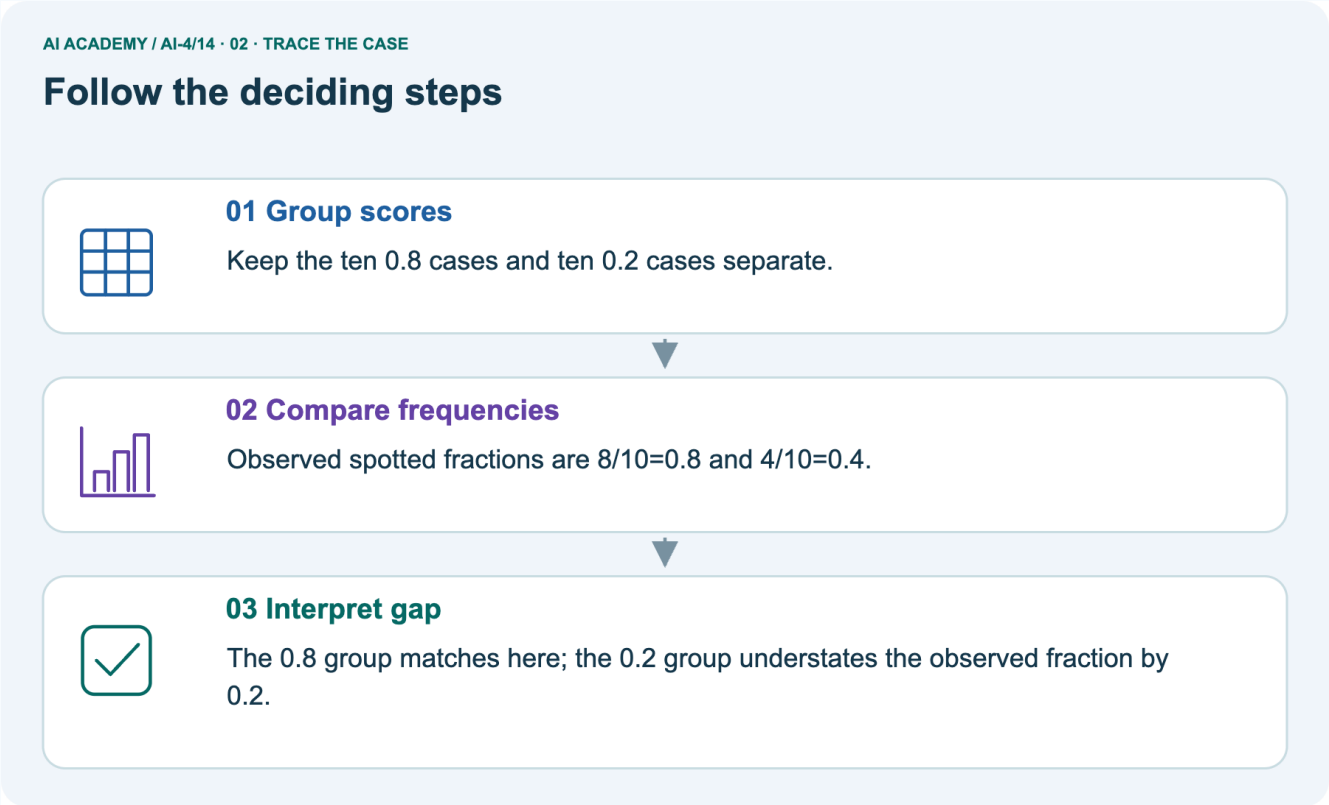
Follow the labelled picture with the matching explanation. Open the vector diagram

Treat **calibration** as a comparison across groups of probability **score**s. Supplied facts: Toy classifier **output**s p(spotted)=0.8 on ten cases; eight are actually spotted. Another ten have p=0.2; four are spotted.

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

> Point to the given fact that matters most. Explain why it matters.

02. Follow the deciding steps



Follow the labelled picture with the matching explanation. Open the vector diagram

Group **score**s: Keep the ten 0.8 cases and ten 0.2 cases separate. → Compare frequencies: Observed spotted fractions are 8/10=0.8 and 4/10=0.4. → Interpret gap: The 0.8 group matches here; the 0.2 group understates the observed fraction by 0.2.

> Point to each numbered stage and explain which recorded **input** it uses.

03. Four-step worked evidence for Week 14: Scores and Calibration Intuition

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Four-step worked evidence for Week 14: Scores and Calibration Intuition

Step	Evidence and reasoning
1	Each score estimates the probability that the selected label is correct. Ten predictions have such scores near 0.8.
2	Only six are correct, so observed correctness is 6 ÷ 10 = 60%.
3	The score ranking may still help, but 0.8 should not be stated as an 80% guarantee.
4	Use bins across many cases and design an uncertain zone requiring human review.

Trace these corrected steps. Keep the stated metric, assumptions and interpretation when you change the case.

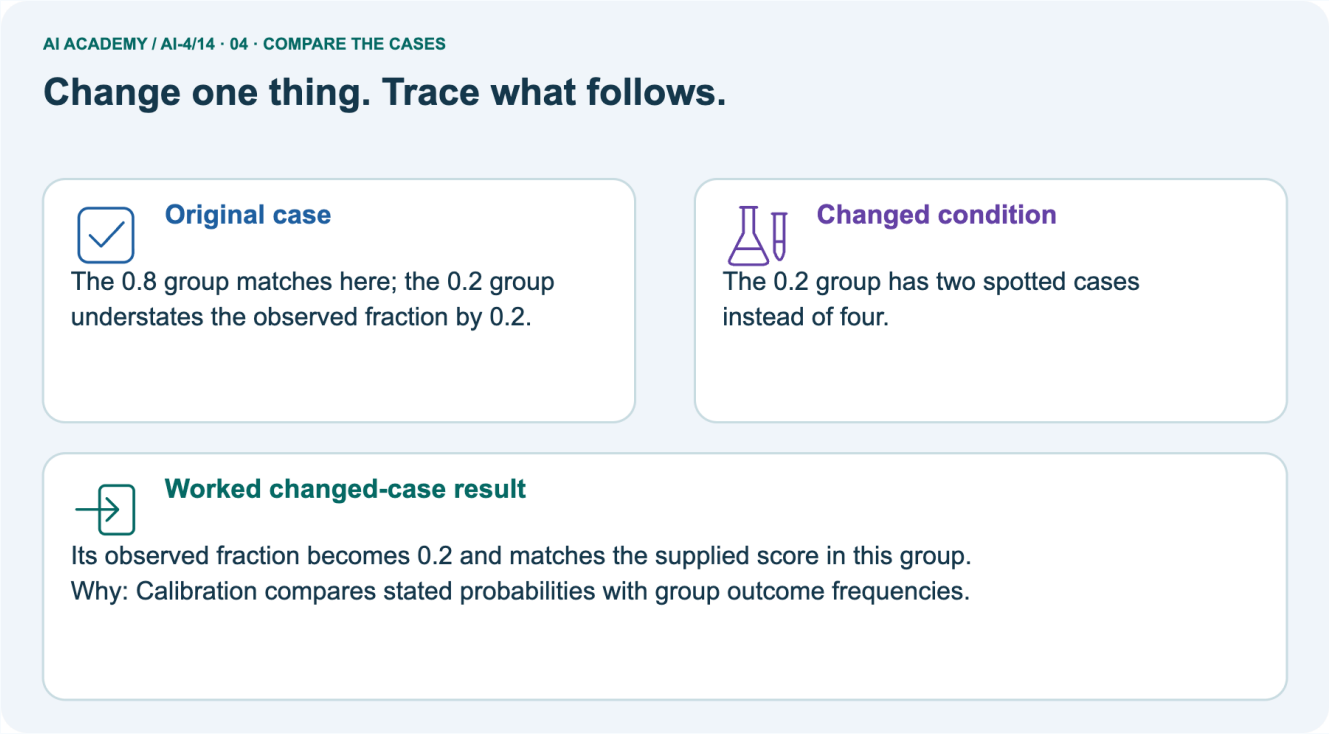
Follow the labelled picture with the matching explanation. Open the vector diagram

Four-step worked **evidence** for Week 14: **Score**s and **Calibration** Intuition. 1 | Each **score** estimates the probability that the selected label is correct.

Ten **prediction**s have such **score**s near 0.8.. 2 | Only six are correct, so observed correctness is 6 ÷ 10 = **60%**.. 3 | The **score** ranking may still help, but 0.8 should not be stated as an **80%** guarantee.. 4 | Use bins across many cases and design an uncertain zone requiring human review.

> Read the labels and values. Explain the relationship the model shows.

04. Change one thing. Trace what follows.



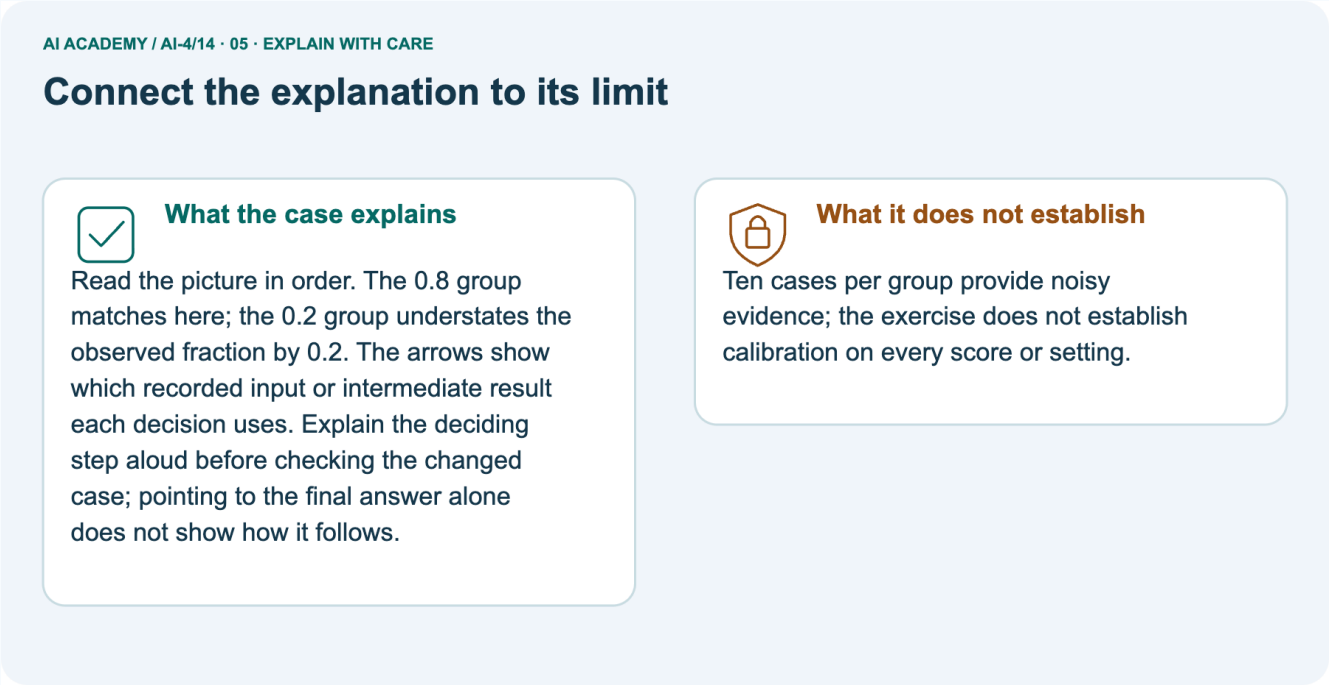
Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: The 0.8 group matches here; the 0.2 group understates the observed fraction by 0.2.
Changed condition: The 0.2 group has two spotted cases instead of four.

Expected result: Its observed fraction becomes 0.2 and matches the supplied **score** in this group.
Reason: **Calibration** compares stated probabilities with group outcome frequencies.

> Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

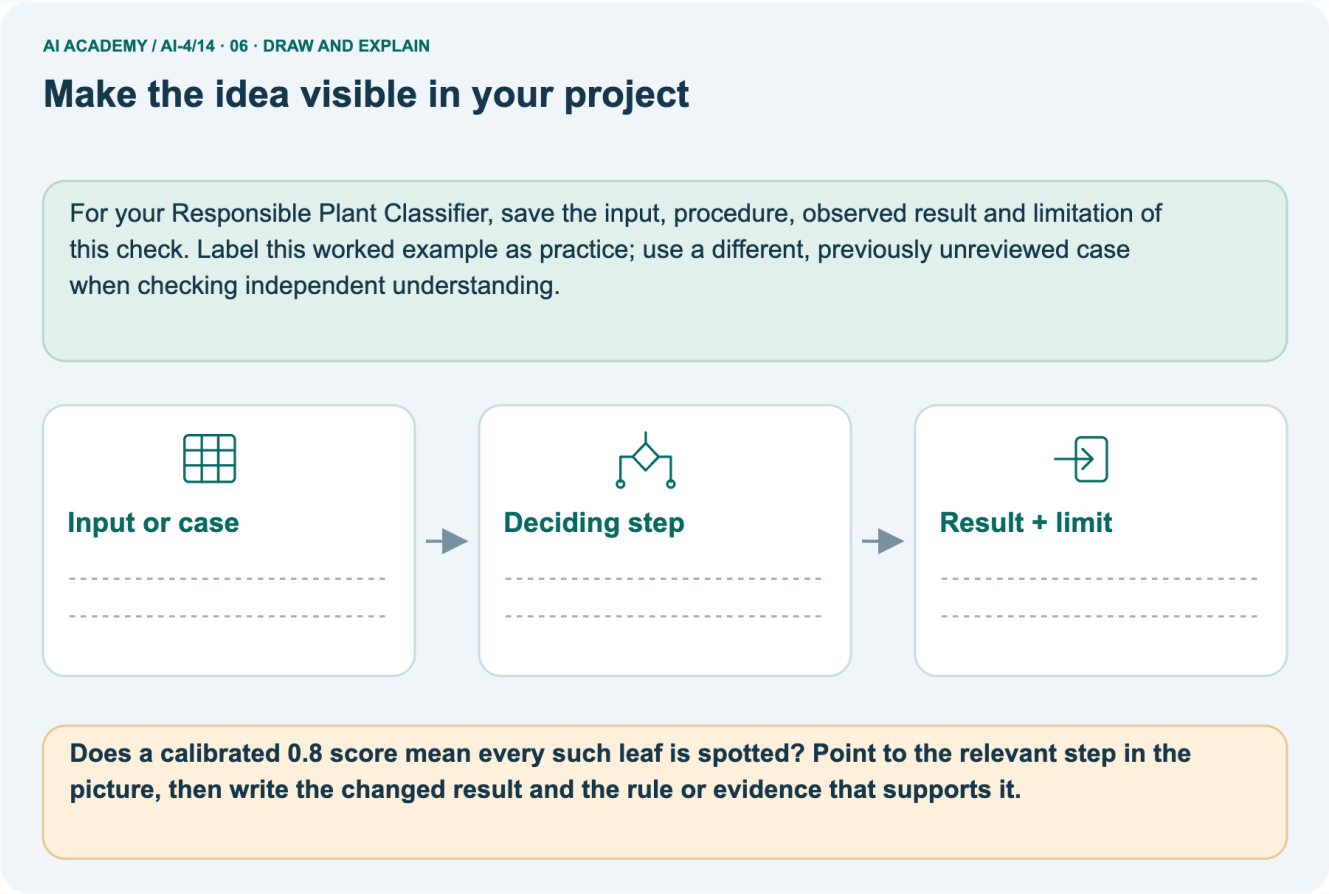
Read the picture in order. The 0.8 group matches here; the 0.2 group understates the observed fraction by 0.2. The arrows show which recorded **input** or intermediate result each decision uses.

Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Limit: Ten cases per group provide noisy **evidence**; the exercise does not establish **calibration** on every **score** or setting.

> Explain the case, then point to a limit on the claim.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For your Responsible Plant Classifier, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding. Draw your **input**, deciding step, result and limit.

Supported question: Does a calibrated 0.8 **score** mean every such leaf is spotted? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

> Draw your project trace and justify the deciding step using **evidence**.

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.